Annex I. Joint IMF/World Bank Debt Sustainability Analysis 2007 under the Debt Sustainability Framework for Low Income Countries[1]

External and public debt dynamics are assessed using the Low Income Country Debt Sustainability Analysis (LIC DSA) framework.[2] *Fund and Bank staffs have concluded that Zambia's* ***risk of external debt distress is low. Total public debt is sustainable under all scenarios****. Zambia's debt sustainability outlook has strengthened substantially since 2005, mainly because of higher copper export receipts and debt relief received under the HIPC and MDRI Initiatives. The baseline scenario shows that Zambia's public debt will improve further over the medium term as long as economic policies remain appropriate. Debt ratios are expected to continue to be manageable, well below the risk thresholds, through 2027.*

I. Underlying Macroeconomic Assumptions

1. **The medium-term framework assumes that the Zambian economy will continue to perform well**. The baseline scenario assumes that real GDP growth remains strong, reflecting large-scale investments in mining and other sectors. The current account deficit is expected to widen over the long run as copper prices fall. Fiscal policies are assumed to remain targeted on limiting domestic borrowing to about one percent of GDP annually. The key assumptions are summarized below. The assumptions for this LIC DSA are broadly in line with those underlying the 2005 LIC DSA, except for GDP growth. The more positive copper sector developments in 2006 and 2007 have been incorporated. While stronger growth is assumed in the medium term, a more conservative approach is taken over the long term.

2. **The baseline scenario is based on the following assumptions:**

- **Economic growth:** Real GDP would continue to grow at 6–6½ percent a year through 2012, supported primarily by developments in the copper and construction sectors. Over time, however, the Zambian economy is expected to diversify as noncopper sectors expand in response to an improved business environment and infrastructure. In the long run Zambia's growth is expected to stabilize at 4½ percent.

- **Inflation:** Inflation should decline to about 5 percent, if monetary and fiscal policies continue to be prudent.

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[1] Prepared by the IMF and World Bank staffs.

[2] The previous LIC DSA (IMF Country Report No. 06/39) showed that Zambia was at a low risk of debt distress. At that time, Zambia had reached HIPC completion but relief under the MDRI has not yet been delivered.

- **External sector:** Copper export volumes would grow at an average annual rate of about 10 percent through 2012 and then 4½ percent through 2027. While copper will remain Zambia's most important export, its share in total exports is expected to fall from 78 percent in 2006 to 67 percent by 2012 and then to about 63 percent by 2027.[3] Imports are assumed to grow in line with GDP in the long run. Annual foreign direct investment would peak at 7 percent of GDP in 2007 and decline to an average of 2½ percent for 2008–12 and to 1½ percent of GDP over the long run.

- **Government revenue and expenditure**: Government revenues would rise gradually from 18.4 percent of GDP in 2007 to 20 percent by 2027, mainly from improvements in tax administration; no change in the tax system is assumed. Expenditures stay constant at about 25 percent of GDP.

- **Financing:** Discussions with the authorities and main donors indicate that there are no plans currently to scale up foreign financing (grants and loans), except for one donor. The baseline scenario therefore assumes that annual external support will remain at about 6 percent of GDP during 2007-12, and then decline to 4½ percent of GDP during 2013-17 and to 3 percent of GDP by 2027.[4] The baseline assumes support from emerging creditors, albeit at low levels. Annual external borrowing would remain at about 1 percent of GDP through 2012 and then decline to about ¾ percent through 2017 and ½ percent for the rest of the period. Domestic financing would fall to 1 percent of GDP by 2009 and remain there over the long term. These funds would be used mainly to finance infrastructure projects.

- **Contingent liabilities:** Public Service Pension Fund's liabilities would grow by ½ percent of GDP per year.

II. External Debt Sustainability Analysis

Background

3. **As a result of the HIPC and MDRI Initiatives, Zambia's stock of external debt has declined substantially.[5]** The external debt-to-GDP ratio declined from about 85 percent

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[3] The nominal growth of exports in U.S. dollar terms is negative for 2009–11 because the price of copper is assumed to decline at an average annual rate of 20 percent over the next three years and at lower rates through 2013, in line with WEO projections. It is assumed that the price of copper will stabilize after 2013.

[4] In the baseline scenario a sizeable portion of grants are allocated to finance investment projects and classified as capital grants.

[5] Zambia reached the completion point under the Enhanced HIPC Initiative in April 2005. MDRI relief was delivered in January 2006.

at the end of 2005 to about 9 percent at the end of 2006. Several creditors have already delivered their share of debt relief.[6]

- **Multilateral Creditors**: All multilateral creditors have provided HIPC relief. The AfDF, IDA, and the Fund have fully delivered their share of MDRI relief.

- **Paris Club Creditors**: Brazil, a creditor invited to participate in the Paris Club meeting in which relief beyond HIPC was agreed, has not yet delivered the relief expected, but is seeking approval from its Senate to do so. Only one full member of the Paris Club (Russia) has not granted beyond HIPC relief to Zambia, but discussions have started.[7]

- **Non-Paris Club Creditors**: Of the nine non-Paris Club creditors, China and India have provided relief on claims held by their governments but not on those held by public enterprises—Zambia plans to contact them. Two creditors, Bulgaria and Iraq, have not yet provided HIPC relief, but the authorities have contacted them. Three non-Paris Club creditors (Kuwait, Romania, and Yugoslavia) sold their claims on Zambia to private creditors and have been fully repaid. Debt to the other two non-Paris Club creditors was fully repaid without any HIPC relief.

Zambia: Public External Debt

	(In percent of GDP)		(In percent of exports of goods and services)	
	2005	2006	2005	2006
Gross external debt	86.0	8.8	251.9	23.3
Medium- and long-term debt	86.0	8.8	251.9	23.3
Multilateral	49.1	5.5	251.9	23.3
IMF	8.1	0.4	143.8	14.6
Other	41.0	5.1	23.8	1.0
Bilateral official	35.9	2.7	120.0	13.6
Paris Club	32.8	1.9	105.1	7.3
Other	3.0	0.9	96.2	5.0
Suppliers and other	1.0	0.6	3.0	1.5

Sources: Bank of Zambia; and staff estimates.

[6] Arrears to private creditors, which accumulated because of limitations on the availability of foreign exchange prior to 1985, have been excluded, since verification of outstanding claims to private individuals is not completed.

[7] Russia is working with the World Bank to develop a debt-for-development swap. This analysis assumes that Russia has already delivered debt relief beyond HIPC.

Results of the External DSA

4. **External debt burden indicators have improved substantially and are expected to remain well below the risk thresholds over the projection period in the baseline scenario** (Table 1). Zambia's external debt ratios remain well below the risk thresholds.[8] The stock of debt is expected to increase to about 12 percent of GDP through 2011 and then decline to single digits by 2017. The NPV of debt-to-GDP is expected to climb from about 6 percent in 2007 to about 7 percent by 2011 and then decline to about 4 percent—below the 40 percent threshold. As copper prices decline and production slows, the NPV of debt-to-exports is projected to increase from 14 percent in 2007 to about 23 percent in 2012 and stay there through 2027, well below the 150 percent threshold. The debt service-to-exports ratio is expected to increase from 1 percent in 2007 to 2 percent by 2012 and to 2.2 percent by 2027.

5. **The standard sensitivity analysis points to a much reduced risk of debt distress (Table 2).** Zambia's external debt position has become more resilient to exogenous shocks. In the alternative scenarios, the ratio that deteriorates most is the NPV of debt-to-exports. This occurs when key variables are assumed to grow at their historical averages (1997–2006), which reflect the poor performance of the Zambian economy from 1997 through 2002. The external current account deficit would remain at 10.7 percent of GDP from 2008 to 2027—significantly higher than the deficit assumed in the baseline scenario (3 percent by 2020). As a result, the NPV of debt-to-exports rises to 45 percent by 2017 and to 106 percent by 2027. All bound tests result in higher but stable ratios well below the thresholds. Thus, even under difficult circumstances, Zambia's risk of debt distress is low. The scenario based on historical averages points to the need to diversify Zambia's exports away from copper as a means to make it even more resilient to external shocks.

6. **Plans for substantial investment in power generation have been under discussion for a long time (Table 1).** Therefore, the 2005 LIC DSA included an alternative scenario of nonconcessional foreign borrowing (US$1 billion for 2008–11).[9] The update assumes construction of three new hydropower plants (Itezhi Tezhi, Kariba North, and Lower Kafue Gorge) during 2009–15, the financing of which requires US$1.2 billion. However, slightly more than 40 percent is expected to be financed with foreign direct investment, reducing the nonconcessional borrowing required to about US$690 million.[10] The non-concessional borrowing will not significantly change the external debt outlook. During the construction period, imports are expected to increase, but as projects are completed exports of electricity

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[8] The World Bank's 2006 Country Policy and Institutions Assessment (CPIA) based on the three-year average ranks Zambia as a medium performer. Thus, the external debt burden thresholds for Zambia are (i) NPV of debt-to-GDP, 40 percent; (ii) NPV of debt-to-exports, 150 percent; (iii) NPV of debt-to-revenue, 250 percent; and (iv) debt service to exports, 20 percent, and to revenue, 30 percent.

[9] This scenario assumed an 8 percent interest rate, 5-year grace period, and 17-year maturity.

[10] This scenario assumes a 6 percent interest rate, 5-year grace period, and 15-year maturity.

will come on stream. The NPV of debt-to-export ratio will peak at 37 percent in 2012 (23 percent in the baseline scenario) but will converge to the baseline scenario by 2027. The assumed borrowing on nonconcessional terms does not pose any threat to the external debt sustainability; debt indicators even under the bound test are below the thresholds.

III. Public Debt Sustainability Assessment

Background

7. **The size and composition of Zambia's public debt have significantly changed.** At end 2005, total public debt was about 105 percent of GDP, 18 percent of which was domestic debt. After extensive debt relief from foreign creditors, Zambia's public debt decreased to 27 percent of GDP and domestic debt's share increased to about 70 percent.

Zambia: Structure of Public Sector Debt

	2005	2006
Public debt	100.0	100.0
External debt	81.8	32.2
Domestic debt	18.2	67.8

Sources: Bank of Zambia; and staff estimates.

8. As of end 2006, the domestic debt had two main components (i) government securities (15.9 percent of GDP), which are denominated in local currency only; and (ii) domestic arrears (2.6 percent of GDP). Most securities are marketable (13 percent of GDP), however, there is no significant secondary market. Foreign investment in government securities began in early 2005, after Zambia reached its HIPC completion point. The estimated stock of government securities in the hands of foreigners has increased from about US$150 million at end-2005 to about US$200 million by end-September 2007 (about 13 percent of the total domestic debt), of which close to 70 percent is short-term debt. This indicates that the rollover risk is low and the possible pressure on reserves of a change in sentiment of foreign investors is moderate.

Results of the Public DSA

9. **In the baseline scenario, Zambia's public debt is expected to decline over the projection period (Table 3).** The NPV of domestic debt would decline from 18.6 percent of GDP in 2006 to 10.4 percent by 2027. As a result of this and the decline in external debt, the NPV of public sector debt-to-GDP would drop from about 27 percent to 16 percent by 2027. Thus, Zambia's public debt is considered manageable, as long as the authorities implement a cautious debt policy.

10. **The standard sensitivity analysis shows that Zambia's public debt is also more resilient to shocks.** The alternative scenarios and bound tests show that the outlook for public debt sustainability is benign, except in the case of permanently lower GDP. In this scenario the NPV of debt-to-revenue rises from about 97 percent to about 139 percent. In the

bound tests, the NPV of debt-to-revenue ratio climbs to about 194 percent by 2027, which underscores the importance of sound macroeconomic policies to achieve higher GDP growth.

IV. CONCLUSIONS

11. **Application of the LIC-DSA framework indicates that Zambia's debt sustainability outlook has strengthened substantially.** Fund and Bank staffs consider that Zambia is at low risk of external debt distress. The external debt indicators remain well below the thresholds in the alternative scenarios and the bound stress tests. The risks to public debt are also low.

Table 1. Zambia: External Debt Sustainability Framework, Baseline Scenario, 2005-27 [1]
(In percent of GDP, unless otherwise indicated)

	Actual		Historical	Standard	Projections									
	2005	2006	Average 6/	Deviation 6/	2007	2008	2009	2010	2011	2012	2007-12 Average	2017	2027	2013-27 Average
External debt (nominal) [1]	**86.0**	**8.8**			**9.9**	**10.1**	**10.7**	**11.4**	**11.7**	**11.5**		**9.6**	**5.4**	
o/w public and publicly guaranteed (PPG)	86.0	8.8			9.9	10.1	10.7	11.4	11.7	11.5		9.6	5.4	
Change in external debt	-40.7	-77.2			1.1	0.2	0.6	0.7	0.3	-0.2		-0.4	-0.4	
Identified net debt-creating flows	-26.3	-32.9			-0.3	-0.1	2.6	4.1	5.0	5.1		1.5	2.1	
Non-interest current account deficit	**9.4**	**-0.2**	**11.1**	**4.8**	**7.4**	**4.9**	**5.7**	**6.8**	**7.6**	**7.7**		**3.6**	**3.7**	3.9
Deficit in balance of goods and services	2.5	-8.0			-3.1	-4.1	-0.7	2.2	3.1	3.2		0.3	2.0	
Exports	33.7	37.5			40.7	40.0	36.1	33.0	31.4	30.4		28.0	17.4	
Imports	36.2	29.6			37.6	35.9	35.4	35.1	34.4	33.6		28.3	19.4	
Net current transfers (negative = inflow)	-3.6	-2.4	-4.6	1.2	-2.9	-2.9	-2.5	-2.5	-1.7	-1.4		-1.1	-0.8	-1.0
o/w official	-3.9	-3.1			-3.6	-3.7	-3.4	-3.6	-2.8	-2.5		-1.9	-1.2	
Other current account flows (negative = net inflow)	10.5	10.2			13.4	11.9	8.9	7.1	6.2	5.9		4.4	2.5	
Net FDI (negative = inflow)	**-5.2**	**-4.3**	**-5.5**	**1.9**	**-7.2**	**-4.5**	**-2.5**	**-2.1**	**-2.1**	**-2.1**		**-1.7**	**-1.5**	-1.6
Endogenous debt dynamics [2]	**-30.4**	**-28.4**			**-0.5**	**-0.5**	**-0.5**	**-0.5**	**-0.5**	**-0.5**		**-0.3**	**-0.2**	
Contribution from nominal interest rate	1.5	0.2			0.1	0.1	0.1	0.1	0.1	0.1		0.2	0.1	
Contribution from real GDP growth	-5.0	-3.6			-0.5	-0.6	-0.6	-0.6	-0.7	-0.6		-0.5	-0.2	
Contribution from price and exchange rate changes	-27.0	-25.0			…	…	…	…	…	…		…	…	
Residual (3-4) [3]	**-14.5**	**-44.3**			**1.4**	**0.3**	**-2.1**	**-3.4**	**-4.7**	**-5.3**		**-1.9**	**-2.4**	
o/w exceptional financing	-6.6	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
NPV of external debt [4]	...	6.0			5.6	5.7	6.1	6.6	7.0	6.9		6.4	3.9	
In percent of exports	...	16.0			13.7	14.3	17.0	20.1	22.2	22.8		22.7	22.5	
NPV of PPG external debt	**...**	**6.0**			**5.6**	**5.7**	**6.1**	**6.6**	**7.0**	**6.9**		**6.4**	**3.9**	
In percent of exports	**...**	**16.0**			**13.7**	**14.3**	**17.0**	**20.1**	**22.2**	**22.8**		**22.7**	**22.5**	
In percent of government revenues	**...**	**35.6**			**31.5**	**32.6**	**35.1**	**37.0**	**38.0**	**37.6**		**34.3**	**21.0**	
Debt service-to-exports ratio (in percent)	**97.1**	**60.4**			**1.3**	**0.9**	**0.9**	**1.0**	**1.4**	**1.9**		**1.6**	**2.2**	
PPG debt service-to-exports ratio (in percent)	**97.1**	**60.4**			**1.3**	**0.9**	**0.9**	**1.0**	**1.4**	**1.9**		**1.6**	**2.2**	
PPG debt service-to-revenue ratio (in percent)	**188.3**	**134.3**			**3.0**	**2.1**	**1.9**	**1.8**	**2.4**	**3.1**		**2.4**	**2.0**	
Total gross financing need (billions of U.S. dollars)	2682.9	1981.7			-225.4	-88.0	234.8	444.8	748.7	1035.0		373.4	1112.1	
Non-interest current account deficit that stabilizes debt ratio	50.1	77.0			6.3	4.7	5.2	6.1	7.3	7.9		3.9	4.0	
Key macroeconomic assumptions														
Real GDP growth (in percent)	5.2	6.2	3.7	2.3	6.2	6.5	6.5	6.0	6.0	6.0	6.2	5.0	4.5	4.7
GDP deflator in US dollar terms (change in percent)	27.0	41.0	9.9	16.6	-2.7	-0.9	-3.6	-4.2	-1.5	1.6	-1.9	3.6	4.1	3.9
Effective interest rate (percent) [5]	1.6	0.3	2.0	...	0.7	1.2	1.0	0.8	1.1	1.1	1.0	1.7	1.6	1.6
Growth of exports of G&S (US dollar terms, in percent)	19.6	66.8	17.3	29.5	12.0	3.7	-7.1	-7.3	-0.7	4.3	0.8	6.3	2.8	4.8
Growth of imports of G&S (US dollar terms, in percent)	13.5	22.4	9.5	13.9	31.5	0.6	1.5	0.8	2.3	5.1	7.0	5.5	4.9	4.9
Grant element of new public sector borrowing (in percent)	...	...	...	...	34.6	37.8	34.8	36.1	33.1	33.2	34.9	31.8	29.2	30.8
Aid flows (in billions of US dollars) [7]	1,075	572			756	709	718	757	681	683		820	1,339	
o/w Grants	0.4	0.5			0.5	0.6	0.6	0.6	0.5	0.5		0.7	1.2	
o/w Concessional loans	176.4	115.7			207.5	115.8	130.3	138.2	143.9	151.9		155.0	184.5	
Grant-equivalent financing (in percent of GDP) [8]	...	...			0.6	0.4	0.4	0.4	0.4	0.4		0.2	0.1	0.2
Grant-equivalent financing (in percent of external financing) 8/	...	...			34.8	38.1	35.1	36.4	33.4	33.4		32.1	29.6	31.2
Memorandum items:														
Nominal GDP (billions of US dollars)	7,271	10,893			11,261	11,884	12,212	12,408	12,956	13,949		21,263	49,409	
(NPVt-NPVt-1)/GDPt-1 (in percent)					-0.2	0.5	0.6	0.6	0.6	0.5	0.4	0.4	0.1	0.3

Source: Staff simulations.

[1]Includes both public and private sector external debt.

[2/]Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

[3/]Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. It also includes grants allocated to finance investment projects, which are classified as capital inflows in the balance of payments. For projections also includes contribution from price and exchange rate changes.

[4] Assumes that NPV of private sector debt is equivalent to its face value.

[5]Current-year interest payments divided by previous period debt stock.

[6]Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

[7]Defined as grants, concessional loans, and debt relief.

[8]Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the NPV of new debt).

Table 2. Zambia: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2007-27
(In percent)

	Projections							
	2007	2008	2009	2010	2011	2012	**2017**	2027
NPV of debt-to-GDP ratio								
Baseline	6	6	6	7	7	7	**6**	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2008-27 [1]	6	9	10	10	9	8	**13**	18
A2. New public sector loans on less favorable terms in 2008-27 [2]	6	6	7	7	8	8	**8**	6
A3. Non-concessional borrowing for power generation plants	6	6	7	9	10	11	**9**	4
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008-09	6	6	7	7	8	8	**7**	4
B2. Export value growth at historical average minus one standard deviation in 2008-09 [3]	6	10	15	16	17	16	**12**	5
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008-09	6	6	7	7	8	8	**7**	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2008-09 [4]	6	6	5	5	6	6	**6**	4
B5. Combination of B1-B4 using one-half standard deviation shocks	6	5	2	3	3	3	**4**	3
B6. One-time 30 percent nominal depreciation relative to the baseline in 2008 [5]	6	8	9	10	10	10	**9**	6
NPV of debt-to-exports ratio								
Baseline	14	14	17	20	22	23	**23**	22
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-26 [1]	14	22	27	29	28	26	**45**	106
A2. New public sector loans on less favorable terms in 2007-26 2/	14	15	18	22	25	27	**29**	35
A3. Non-concessional borrowing for power generation plants	14	14	19	27	33	37	**32**	22
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008-09	14	14	17	20	22	23	**23**	22
B2. Export value growth at historical average minus one standard deviation in 2008-09 [3]	14	30	54	62	66	67	**55**	39
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008-09	14	14	17	20	22	23	**23**	22
B4. Net non-debt creating flows at historical average minus one standard deviation in 2008-09 [4]	14	15	14	17	19	19	**20**	21
B5. Combination of B1-B4 using one-half standard deviation shocks	14	13	6	8	9	10	**13**	17
B6. One-time 30 percent nominal depreciation relative to the baseline in 2008 [5]	14	14	17	20	22	23	**23**	22
Debt service-to-exports ratio								
Baseline	1	1	1	1	1	2	**2**	2
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2008-27 [1]	1	1	1	1	1	2	**2**	6
A2. New public sector loans on less favorable terms in 2008-27 [2]	1	1	1	1	2	2	**2**	3
A3. Non-concessional borrowing for power generation plants	1	1	1	1	2	3	**3**	2
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008-09	1	1	1	1	1	2	**1**	2
B2. Export value growth at historical average minus one standard deviation in 2008-09 [3]	1	1	2	3	3	3	**4**	4
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008-09	1	1	1	1	1	2	**1**	2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2008-09 [4]	1	1	1	1	1	2	**1**	2
B5. Combination of B1-B4 using one-half standard deviation shocks	1	1	1	1	1	1	**1**	1
B6. One-time 30 percent nominal depreciation relative to the baseline in 2008 [5]	1	1	1	1	1	2	**1**	2
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) [6]	30	30	30	30	30	30	**30**	30

Source: Staff projections and simulations.

[1]Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

[2]Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

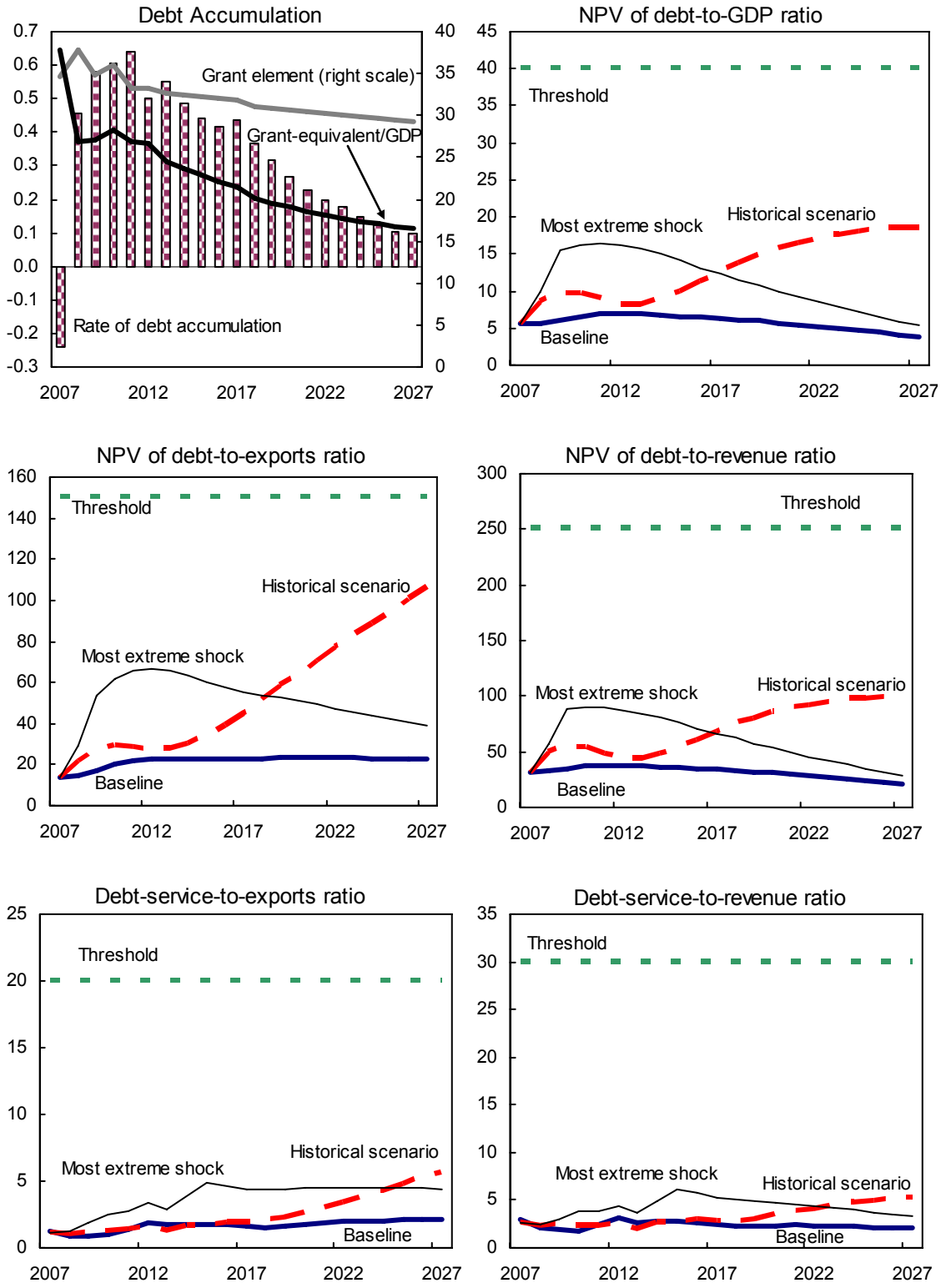
[3]Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly a
an offsetting adjustment in import levels).

[4]Includes official and private transfers and FDI.

[5]Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

[6]Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Figure 1. Zambia: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2007-2027



Source: Staff projections and simulations.

Table 3.Zambia: Public Sector Debt Sustainability Framework, Baseline Scenario, 2005-27
(In percent of GDP, unless otherwise indicated)

	Actual				Estimate				Projections				
	2005	2006	Historical Average[5]	Standard Deviation[5]	2007	2008	2009	2010	2011	2012	2007-12 Average	2017	2027
Public sector debt [1]	105.2	27.4			26.6	25.3	24.9	25.5	25.6	25.1		21.6	15.8
o/w foreign-currency denominated	86.0	8.8			9.9	10.1	10.7	11.4	11.7	11.5		9.6	5.4
Change in public sector debt	-43.4	-77.8			-0.7	-1.3	-0.4	0.6	0.0	-0.5		-0.6	-0.5
Identified debt-creating flows	-55.4	4.0			-2.4	0.1	0.6	0.8	0.6	0.1		0.6	1.0
Primary deficit	0.1	0.9	1.5	1.6	-0.2	0.5	0.4	0.5	0.9	0.9	0.5	0.7	0.9
Revenue and grants	23.0	21.5			23.2	23.0	22.8	23.0	23.2	23.0		22.6	22.3
of which : grants	5.7	4.2			4.9	5.0	4.8	5.0	4.1	3.8		3.1	2.3
Primary (noninterest) expenditure	23.1	22.4			23.0	23.5	23.2	23.5	24.2	23.9		23.3	23.2
Automatic debt dynamics	-54.2	3.1			-2.6	-1.0	-0.3	-0.3	-0.8	-1.3		-0.7	-0.4
Contribution from interest rate/growth differential	-11.6	-7.9			-1.7	-1.2	-0.9	-0.9	-1.1	-1.1		-0.6	-0.3
of which : contribution from average real interest rate	-4.0	-2.6			-0.1	0.3	0.7	0.7	0.4	0.4		0.5	0.4
of which : contribution from real GDP growth	-7.6	-5.2			-1.6	-1.6	-1.6	-1.5	-1.5	-1.4		-1.1	-0.7
Contribution from real exchange rate depreciation	-42.6	10.9			-0.9	0.3	0.6	0.6	0.3	-0.2		...	...
Other identified debt-creating flows	-1.2	0.0			0.5	0.5	0.5	0.5	0.5	0.5		0.5	0.5
Privatization receipts (negative)	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0
Recognition of implicit or contingent liabilities (pensions)	0.0	0.0			0.5	0.5	0.5	0.5	0.5	0.5		0.5	0.5
Debt relief (HIPC and other)	-1.5	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0
Other (specify, e.g. bank recapitalization)	0.2	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0
Residual, including asset changes	12.0	-81.8			1.6	-1.4	-1.0	-0.2	-0.6	-0.6		-1.2	-1.5
NPV of public sector debt	40.8	25.9			22.5	21.1	20.6	21.0	21.0	20.5		18.4	14.4
o/w foreign-currency denominated	21.6	7.4			5.8	5.9	6.4	6.8	7.1	7.0		6.5	4.0
o/w external	21.6	7.4			5.8	5.9	6.4	6.8	7.1	7.0		6.5	4.0
Gross financing need [2]	32.5	35.1			5.8	6.6	6.1	5.9	6.2	6.2		5.6	5.1
NPV of public sector debt-to-revenue and grants ratio (in percent)	177.1	120.7			96.8	91.8	90.5	91.3	90.3	89.3		81.6	64.7
NPV of public sector debt-to-revenue ratio (in percent)	236.2	150.0			122.5	117.3	114.6	116.6	109.9	106.9		94.7	72.3
o/w external [3]	125.2	42.6			31.4	32.8	35.3	38.0	37.3	36.2		33.3	20.0
Debt service-to-revenue and grants ratio (in percent) [4]	118.3	137.3			9.2	7.0	6.8	6.1	6.0	7.0		6.9	6.3
Debt service-to-revenue ratio (in percent) [4]	157.7	170.6			11.7	8.9	8.6	7.9	7.3	8.4		8.0	7.0
Primary deficit that stabilizes the debt-to-GDP ratio	43.5	78.7			0.5	1.9	0.8	0.0	0.9	1.4		1.4	1.4
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	5.4	5.2	3.1	2.4	6.2	6.2	6.5	6.5	6.0	6.0	6.3	5.0	4.5
Average real interest rate on domestic currency debt (in percent)	-8.7	-2.6	-15.4	10.3	0.1	2.5	5.3	5.8	3.4	3.4	3.4	3.6	3.8
Real exchange rate depreciation (in percent, + indicates depreciation)	-36.2	13.7	-6.4	19.2	-11.5	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	18.7	13.8	21.0	4.7	10.0	5.5	3.2	2.3	3.8	4.6	4.9	6.2	6.7
Growth of real primary spending (deflated by GDP deflator, in percent)	4.7	3.1	2.2	6.9	9.2	8.8	5.3	7.5	8.9	4.1	7.3	4.9	4.5

Sources: Country authorities; and Fund staff estimates and projections.

[1] Central government.

[2] Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

[3] Revenues excluding grants.

[4] Debt service is defined as the sum of interest and amortization of medium and long-term debt.

[5] Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability and excluding one-off impact of MDRI relief on grants in 2006.

Table 4. Zambia: Sensitivity Analysis for Key Indicators of Public Debt 2007-27

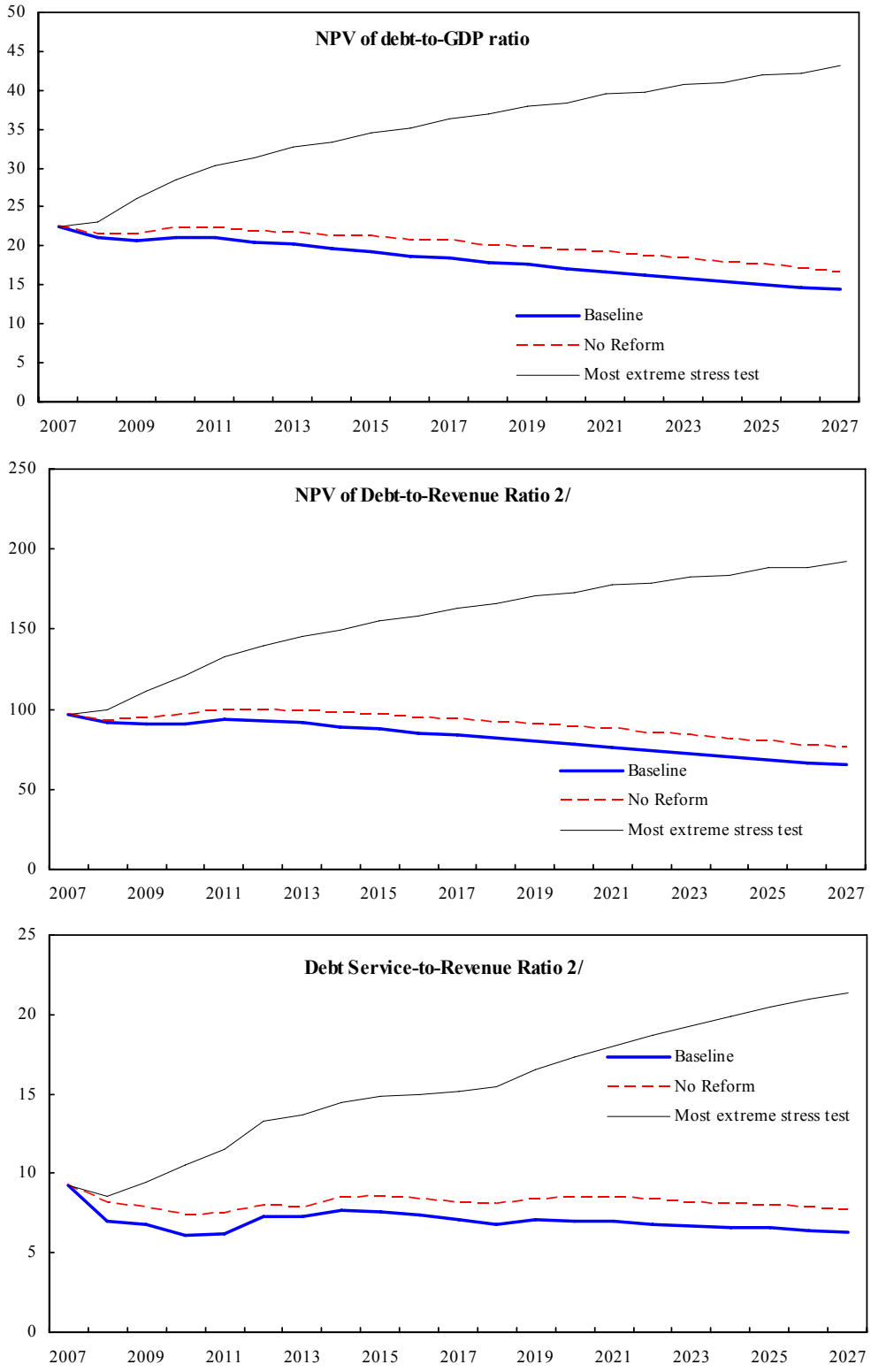
	Projections							
	2007	2008	2009	2010	2011	2012	2017	2027
NPV of Debt-to-GDP Ratio								
Baseline	22	21	21	21	21	21	18	14
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	22	23	24	26	27	28	30	29
A2. Primary balance is unchanged from 2006	22	21	22	22	22	22	21	17
A3. Permanently lower GDP growth [1]	22	21	21	22	23	22	24	31
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2008-2009	22	23	26	28	30	32	37	44
B2. Primary balance is at historical average minus one standard deviations in 2008-2009	22	24	26	26	26	25	22	16
B3. Combination of B1-B2 using one half standard deviation shocks	22	24	26	26	26	25	20	12
B4. One-time 30 percent real depreciation in 2008	22	24	23	24	23	23	20	15
B5. 10 percent of GDP increase in other debt-creating flows in 2008	22	30	30	30	30	30	26	20
NPV of Debt-to-Revenue Ratio [2]								
Baseline	97	92	90	91	90	89	82	65
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	97	98	103	111	114	118	128	125
A2. Primary balance is unchanged from 2006	97	93	94	97	96	95	91	75
A3. Permanently lower GDP growth [1]	97	93	93	96	97	97	104	139
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2008-2009	97	100	112	121	128	135	161	194
B2. Primary balance is at historical average minus one standard deviations in 2008-2009	97	102	113	114	112	110	97	71
B3. Combination of B1-B2 using one half standard deviation shocks	97	102	112	111	108	105	87	54
B4. One-time 30 percent real depreciation in 2008	97	103	102	103	101	99	88	67
B5. 10 percent of GDP increase in other debt-creating flows in 2008	97	132	131	132	131	129	116	88
Debt Service-to-Revenue Ratio [2]								
Baseline	9	7	7	6	6	7	7	6
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	9	8	9	9	9	10	11	13
A2. Primary balance is unchanged from 2006	9	8	8	7	7	8	8	8
A3. Permanently lower GDP growth [1]	9	8	8	7	7	8	10	15
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2008-2009	9	9	9	11	11	13	15	22
B2. Primary balance is at historical average minus one standard deviations in 2008-2009	9	8	11	12	9	9	8	8
B3. Combination of B1-B2 using one half standard deviation shocks	9	8	10	11	8	8	7	5
B4. One-time 30 percent real depreciation in 2008	9	8	8	7	7	8	7	6
B5. 10 percent of GDP increase in other debt-creating flows in 2008	9	8	21	12	9	10	9	10

Sources: Country authorities; and Fund staff estimates and projections.

[1] Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of 20 (i.e., the length of the projection period).

[2] Revenues are defined inclusive of grants.

Figure 2. Zambia: Indicators of Public Debt under Alternative Scenarios, 2007–27[1]



Source: Staff projections and simulations.
1/ Most extreme stress test is test that yields highest ratio in 2017.
2/ Revenue including grants.